

# LOW MOLECULAR WEIGHT NBR: PATENT LANDSCAPE AND PROCESS-CONTROL SYNTHESIS REVIEW

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## 1. ABSTRACT

This report reviews the provided patent set against the target of low molecular weight NBR (nitrile-butadiene rubber) synthesis and process control. The manifested patents do not directly disclose NBR preparation routes; instead, they illustrate adjacent polymerization and manufacturing strategies relevant to controlling molecular weight, viscosity, architecture, and product handling in polymer systems. US10167348B2 is the most mechanistically relevant, describing solution anionic polymerization of 1,1-disubstituted alkene monomers with explicit control over initiator ratio, temperature, solvent choice, monomer feed, and quench conditions, producing homopolymers, random copolymers, and block copolymers with tunable molecular weight distributions. US7683134B2 demonstrates continuous reactive extrusion maleation of polypropylene, showing how feed ratios, peroxide initiator rate, screw speed, and vacuum can be used to tune molecular weight and grafting efficiency in a melt process. US8470957B2 focuses on continuous microreactor production of monomer-like bischloroformates and low-repeating-unit oligomers, emphasizing flow rate, temperature, and stoichiometric control to suppress side reactions and obtain low-degree oligomeric products. US9434846B2 discloses emulsion polymerization of HASE thickeners with surfactant-resistant rheology, using nitrogen purge, 80°C batch hold, ammonium persulfate initiation, and chaser chemistry to convert residual monomer to oligomer and reduce VOCs. Across the set, the dominant process trends are solution or emulsion polymerization, continuous or semi-continuous feed control, explicit initiator/monomer ratio management, temperature control, and post-reaction quenching or stripping. For low-MW NBR synthesis, the evidence supports prioritizing controlled radical or anionic solution polymerization concepts, tight monomer-to-initiator or chain-transfer control, and solvent/temperature regimes that limit runaway viscosity and broadened molecular-weight distributions.

## 2. METHODOLOGY

### PRIMARY PATENT EVIDENCE

#### Patent 1

- Patent Number: US10167348B2
- Patent Title: Solution polymers formed from methylene malonate monomers, polymerization, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses solution anionic polymerization of 1,1-disubstituted alkene monomers using a weak-base or organolithium activator in solvents such as tetrahydrofuran. The process supports homopolymers, random copolymers, and block copolymers, with molecular weight controlled by monomer-to-initiator ratio, temperature, solvent, and monomer feeding strategy. - Monomer-to-activator ratio: about 1000:1 — Example H-1 - Reaction time: about 1 hour — Example H-1 - Reaction temperature: about 23° C. — Example H-1 - Solvent: tetrahydrofuran — Example H-1 - Monomer mass: about 1.0 g — Example H-1 - Solvent mass: about 9.0 g — Example H-1 - Quench reagent: trifluoroacetic acid (TFA) — Example H-1 - Cold precipitation temperature: 0° C. methanol — Example H-1 - First peak molecular weight: about 2000 — Example H-1 - Second peak molecular weight: about 60,000 — Example H-1 - Polydispersity index: about 1.43 — Example H-1 - Monomer-to-activator ratio: about 1000:1 — Example A-1 - Reaction time: 24 hours — Example A-1 - Reaction temperature: about 23° C. — Example A-1 - Weight average molecular weight: about 405,700 — Example A-1 - Number average molecular weight: about 198,000 — Example A-1 - Reaction temperature: about -78° C. — Example N-1 - Activator concentration: 1.5 M secondary butyl lithium in cyclohexane — Example N-1 - Monomer-to-activator ratio: about 1000:1 — Example N-1 - Reaction time: about 20 minutes — Example N-1 - Mn: 12,750 daltons — Example N-1, 2 minutes aliquot - Mn: 20,980 daltons — Example N-1, 6 minutes aliquot - Mn: 36,940 daltons — Example N-1, 10 minutes aliquot - Mn: 41,280 daltons — Example N-1, 20 minutes aliquot - PDI: 2.2 — Example N-1, 2 minutes aliquot - PDI: 1.8 — Example N-1, 6 minutes aliquot - PDI: 1.2 — Example N-1, 10 minutes aliquot - PDI: 1.2 — Example N-1, 20 minutes aliquot - Initial monomer charge: 0.25 g — Example N-2 - Monomer-to-activator ratio: about 1000:1 — Example N-2 - Mn: 48,000 daltons — Example N-2, 4 minutes - Mn: 75,000 daltons — Example N-2, 6 minutes - Mn: 100,000 daltons — Example N-2, 8 minutes - Mn: 130,000 daltons — Example N-2, 10 minutes - Total monomer-to-activator ratio: about 100:1 — Example N-3 - Mn: 8,000 daltons — Example N-3, 2 minutes - Mn: 22,000 daltons — Example N-3, 4 minutes - Mn: 41,000 daltons — Example N-3, 6 minutes - Mn: 58,000 daltons — Example N-3, 8 minutes - Mn: 81,000 daltons — Example N-3, 10 minutes - Monomer ratio HM3:DEMM: 75:25 — Example R-2 - Monomer ratio HM3:DEMM: 50:50 — Example R-3 - Monomer ratio HM3:DEMM: 25:75 — Example R-4 - Total monomer amount: 1 g — Example R-2/R-3/R-4 - Reaction time: about 1 hour — Example R-2/R-3/R-4 - Reaction temperature: about 23° C. — Example R-2/R-3/R-4 - Monomer-to-activator ratio: about 1000:1 — Example R-1 - Mn: 7,104 daltons — Example R-1 - Mw: 16,343 daltons — Example R-1 - PDI: about 2.3 — Example R-1 - Monomer-to-activator ratio: about 1000:1 — Example B-1 - Reaction

temperature: about -78° C. — Example B-1 - Monomer block charge: about 0.25 g — Example B-1 - Reaction time per block: about 5 minutes — Example B-1 - Mn: 28,156 daltons — Example B-1, 1st aliquot - Mn: 41,147 daltons — Example B-1, 2nd aliquot - Mn: 59,243 daltons — Example B-1, 3rd aliquot - Mn: 67,400 daltons — Example B-1, 4th aliquot - Conversion of monomer to polymer: ≈100% — Example B-1 - Mn: about 2,000,000 daltons — Example S-1 - Mn: about 500,000 daltons — Example S-2 - Mn: about 200,000 daltons — Example S-3 - Mn: about 700,000 daltons — Example S-4 - Mn: about 150,000 daltons — Example S-5 - Monomer-to-activator ratio: about 100:1 — Example P-1 - Monomer-to-activator ratio: about 1000:1 — Example P-2 - Monomer purity: about 94.1 weight percent — Example P-1 - Monomer purity: about 98.23 weight percent — Example P-2 - Mn: about 6,700 daltons — Example P-1 - Mw: about 17,400 daltons — Example P-1 - PDI: about 2.60 — Example P-1 - Tg: about 83° C. — Example P-1 - Mn: about 1,451,800 daltons — Example P-2 - Mw: about 2,239,300 daltons — Example P-2 - PDI: about 1.62 — Example P-2 - Tg: about 145° C. — Example P-2 - Monomer-to-activator ratio: about 100:1 — Example P-3/P-4 - Monomer purity: about 92.8 weight percent — Example P-3 - Monomer purity: about 98.6 weight percent — Example P-4 - Mw: about 40,300 daltons — Example P-3 - Tg: about 136° C. — Example P-3 - Mw: about 290,800 daltons — Example P-4 - Tg: about 190° C. — Example P-4 - Solvent: tetrahydrofuran — Example X-1/X-2/X-3/X-4 - Scale: about 1 g polymer — Example X-1/X-2/X-3 - Scale: 450 g polymer — Example X-4 - Reactor size: 8 L round bottom flask — Example X-4 - Solvent amount: 4.05 kg — Example X-4 - Mixing speed: 500 rpm — Example X-4 - Activator amount: about 0.103 mL pure TMG — Example X-4 - Reaction time: 1 hour — Example X-4 - Temperature rise: about 19° C. over first 15 minutes — Example X-4 - Mn: 502,800 daltons — Example X-1 - Mn: 527,300 daltons — Example X-2 - Mn: 493,700 daltons — Example X-3 - Mn: 528,400 daltons — Example X-4 - PDI: 2.5 — Example X-1 - PDI: 2.4 — Example X-2 - PDI: 2.3 — Example X-3 - PDI: 2.1 — Example X-4 - Conversion of monomer to polymer: ≈100% — Example X-1/X-2/X-3/X-4

**Relevant Experimental Evidence** - Example H-1: Demonstrated solution anionic polymerization at 23° C. with 1000:1 monomer-to-activator ratio and low PDI product. - Example N-1: Showed low-temperature (-78° C.) anionic polymerization with Mn increasing over time and narrowing PDI. - Example N-2: Demonstrated living polymer growth by sequential monomer addition with near-linear Mn increase. - Example B-1: Showed block copolymer formation by alternating monomer feeds with ~100% conversion and narrow PDI. - Example X-4: Demonstrated scale-up to 450 g polymer in an 8 L reactor with maintained high conversion.

**Technical Relevance** This is the most relevant patent in the set because it directly addresses controlled solution polymerization, molecular-weight tuning, and viscosity management through initiator ratio, temperature, solvent, and feed strategy. Although it is not NBR-specific, the disclosed control principles are directly transferable to low-MW nitrile-butadiene copolymer synthesis.

## Patent 2

- Patent Number: US7683134B2
- Patent Title: Maleated high acid number high molecular weight polypropylene of low color

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses continuous melt maleation of polypropylene in a twin-screw extruder using molten maleic anhydride and a peroxide initiator. The process uses controlled feed ratios, screw speed, and vacuum removal to produce maleated polypropylene with targeted acid number, molecular weight, and color. - Extruder temperature: 200° C. — Example 1 - Screw speed: 150 rpm — Example 1 - Polypropylene feed rate: 272 kg per hour — Example 1 - Maleic anhydride feed rate: 10.9 kg per hour — Example 1 - Maleic anhydride temperature: 90° C. — Example 1 - Initiator feed rate: 1.1 kg per hour LUPERSOL 101 — Example 1 - Vacuum: 30 inches of Mercury (760 mm) — Example 1 - Acid number: 8.7 — Example 1 - Mn: 48,000 — Example 1 - Mw: 119,000 — Example 1 - Yellowness index color: 51 — Example 1 - Maleic anhydride utilized: 37% — Example 1 - Screw speed: 292 rpm — Example 2 - Acid number: 10.1 — Example 2 - Mn: 43,000 — Example 2 - Mw: 105,000 — Example 2 - Yellowness index color: 49 — Example 2 - Maleic anhydride used: 43% — Example 2 - Initiator feed rate: 2.4 kg per hour LUPERSOL 101 — Example 3 - Acid number: 16.4 — Example 3 - Mn: 30,000 — Example 3 - Mw: 72,000 — Example 3 - Yellowness index color: 48 — Example 3 - Maleic anhydride used: 70% — Example 3 - Screw speed: 150 rpm — Example 4 - Acid number: 14.6 — Example 4 - Mn: 31,000 — Example 4 - Mw: 87,000 — Example 4 - Yellowness index color: 56 — Example 4 - Maleic anhydride used: 62% — Example 4 - Initiator feed rate: 0.5 kg per hour LUPERSOL 101 — Example 5 - Maleic anhydride feed rate: 4.5 kg per hour — Example 5 - Acid number: 5.9 — Example 5 - Mn: 47,000 — Example 5 - Mw: 118,000 — Example 5 - Yellowness index color: 25 — Example 5 - Maleic anhydride used: 60% — Example 5 - Initiator feed rate: 1.1 kg per hour LUPERSOL 101 — Example 6 - Acid number: 9.1 — Example 6 - Mn: 36,000 — Example 6 - Mw: 89,000 — Example 6 - Yellowness index color: 24 — Example 6 - Maleic anhydride used: 93% — Example 6 - Screw speed: 292 rpm — Example 7 - Acid number: 4.8 — Example 7 - Mn: 51,000 — Example 7 - Mw: 130,000 — Example 7 - Yellowness index color: 33 — Example 7 - Maleic anhydride used: 49% — Example 7 - Initiator feed rate: 0.5 kg per hour LUPERSOL 101 — Example 8 - Acid number: 3.0 — Example 8 - Mn: 57,000 — Example 8 - Mw: 148,000 — Example 8 - Yellowness index color: 32 — Example 8 - Maleic anhydride used: 31% — Example 8 - Initiator feed rate: 0.3 kg per hour LUPERSOL 101 — Example 9 - Acid number: 1.8 — Example 9 - Mn: 65,000 — Example 9 - Mw: 165,000 — Example 9 - Yellowness index color: 33 — Example 9 - Maleic anhydride used: 18% — Example 9 - Screw speed: 150 rpm — Example 10 - Acid number: 3.5 — Example 10 - Mn: 58,000 — Example 10 - Mw: 145,000 — Example 10 - Yellowness index color: 23 — Example 10 - Maleic anhydride used: 36% — Example 10 - Maleic anhydride feed rate: 10.9 kg per hour — Example 11 - Acid number: 5.5 — Example 11 - Mn: 64,000 — Example 11 - Mw: 168,000 — Example 11 - Yellowness index color: 36 — Example 11 - Maleic anhydride used: 23% — Example 11 - Screw speed: 292 rpm — Example 12 - Acid number: 3.6 — Example 12 - Mn: 60,000 — Example 12 -

Mw: 150,000 — Example 12 - Yellowness index color: 47 — Example 12 - Maleic anhydride used: 15% — Example 12 - Initiator feed rate: 0.5 kg per hour LUPERSOL 101 — Example 13 - Acid number: 6.0 — Example 13 - Mn: 63,000 — Example 13 - Mw: 135,000 — Example 13 - Yellowness index color: 44 — Example 13 - Maleic anhydride used: 26% — Example 13 - Screw speed: 200 rpm — Example 14 - Maleic anhydride feed rate: 16.3 kg per hour — Example 14 - Acid number: 5.4 — Example 14 - Mn: 60,000 — Example 14 - Mw: 143,000 — Example 14 - Yellowness index color: 64 — Example 14 - Maleic anhydride used: 15% — Example 14 - Screw speed: 292 rpm — Example 15 - Acid number: 4.2 — Example 15 - Mn: 71,000 — Example 15 - Mw: 190,000 — Example 15 - Yellowness index color: 63 — Example 15 - Maleic anhydride used: 12% — Example 15 - Maleic anhydride feed rate: 21.8 kg per hour — Example 16 - Acid number: 3.7 — Example 16 - Mn: 68,000 — Example 16 - Mw: 192,000 — Example 16 - Yellowness index color: 76 — Example 16 - Maleic anhydride used: 8% — Example 16 - Initiator feed rate: 4.5 kg per hour LUPERSOL 101 — Example 17 - Acid number: 9.0 — Example 17 - Mn: 67,000 — Example 17 - Mw: 177,000 — Example 17 - Yellowness index color: 75 — Example 17 - Maleic anhydride used: 19% — Example 17 - Maleic anhydride feed rate: 16.3 kg per hour — Example 18 - Acid number: 12.0 — Example 18 - Mn: 54,000 — Example 18 - Mw: 129,000 — Example 18 - Yellowness index color: 70 — Example 18 - Maleic anhydride used: 34% — Example 18 - Initiator feed rate: 3.4 kg per hour LUPERSOL 101 — Example 19 - Acid number: 16.9 — Example 19 - Mn: 34,000 — Example 19 - Mw: 82,000 — Example 19 - Yellowness index color: 50 — Example 19 - Maleic anhydride used: 72% — Example 19 - Initiator feed rate: 0.5 kg per hour LUPERSOL 101 — Example 20 - Acid number: 4.0 — Example 20 - Mn: 62,000 — Example 20 - Mw: 159,000 — Example 20 - Yellowness index color: 58 — Example 20 - Maleic anhydride used: 11% — Example 20

**Relevant Experimental Evidence** - Example 1: Established baseline continuous maleation at 200° C., 150 rpm, with 37% maleic anhydride utilization. - Example 3: Increasing initiator rate raised acid number but lowered Mn and Mw. - Example 5: Lower initiator and maleic anhydride feed improved color and maintained moderate molecular weight. - Example 15: Higher screw speed and adjusted feed gave Mn 71,000 and Mw 190,000. - Example 20: Demonstrated that lower initiator feed can preserve higher molecular weight while keeping acid number near 4.

**Technical Relevance** This patent is indirectly relevant because it shows how continuous reactive extrusion and peroxide initiation can be used to tune molecular weight and viscosity in a polymer modification process. The control concepts are useful for understanding how process variables can drive molecular-weight reduction or retention in rubber-like materials, though it does not disclose NBR synthesis.

### Patent 3

- Patent Number: US8470957B2
- Patent Title: Method for producing bischloroformate compound, polycarbonate oligomer having ...
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses continuous production of bischloroformate compounds from dihydric phenols, phosgene, and aliphatic tertiary amines in hydrophobic organic solvent systems, followed by use of the bischloroformate to make low-repeating-unit polycarbonate oligomers. The process emphasizes flow-rate control, temperature control, and suppression of hydrolysis and side reactions to obtain monomer-like oligomeric products. - TEA amount: 59.8 g (0.591 mol) — Example 1-1 - MDC amount: 100 mL — Example 1-1 - 4,4'-biphenol amount: 50.0 g (0.269 mol) — Example 1-1 - Phosgene amount: 80.0 g (0.809 mol) — Example 1-1 - Reaction temperature: 13 to 16° C. — Example 1-1 - Reaction time: 3 hours and 6 minutes — Example 1-1 - Stirring time: 1 hour and 38 minutes — Example 1-1 - TEA amount: 29 g (0.287 mol) — Example 1-2 - MDC amount: 70 mL — Example 1-2 - BPZ amount: 35 g (0.13 mol) — Example 1-2 - Phosgene amount: 38.7 g (0.39 mol) — Example 1-2 - Reaction temperature: 5 to 17° C. — Example 1-2 - Reaction time: 3 hours — Example 1-2 - Stirring time: 1 hour — Example 1-2 - Triphosgene amount: 38.7 g (0.13 mol) — Example 1-3 - MDC amount: 200 mL — Example 1-3 - BPZ amount: 35 g (0.13 mol) — Example 1-3 - Reaction temperature: 3 to 5° C. — Example 1-3 - TEA amount: 29 g (0.287 mol) — Example 1-3 - Reaction temperature: 11 to 18° C. — Example 1-3 - Reaction time: 3 hours — Example 1-3 - Stirring time: 1 hour — Example 1-3 - TEA amount: 223.9 g (2.21 mol) — Example 1-4 - MDC amount: 460 mL — Example 1-4 - Bis-OCF amount: 230 g (1.01 mol) — Example 1-4 - Phosgene amount: 223 g (2.25 mol) — Example 1-4 - Reaction temperature: 15.5 to 19° C. — Example 1-4 - Reaction time: 3 hours and 3 minutes — Example 1-4 - Stirring time: 1 hour — Example 1-4 - TEA amount: 17.0 g (0.168 mol) — Example 1-5 - MDC amount: 35 mL — Example 1-5 - Material solution concentration: 0.20 to 0.30 kg/L — Example 2-2/2-3/2-4 - Reaction temperature: 5 to 11° C. — Experimental Example 3-1 - Reaction time: 2 hours and 46 minutes — Experimental Example 3-1 - Reduced viscosity: 0.94 — Experimental Example 3-1 - Reaction temperature: 25 to 32° C. — Experimental Example 3-2 - Reaction temperature: 4 to 8° C. — Experimental Example 3-3 - Flow rate of material solution: 50 mL/min — Example 4-1 - Flow rate of phosgene solution: 37.2 mL/min — Example 4-1 - Linear velocity of mixture: 5.8 mL/s — Example 4-1 - Phosgene equivalent weight: 1.5 equivalent weight relative to phenolic hydroxyl group — Example 4-1 - Flow rate of material solution: 250 mL/min — Example 4-2 - Flow rate of phosgene solution: 186 mL/min — Example 4-2 - Flow rate of material solution: 200 mL/min — Example 4-3 - Flow rate of phosgene solution: 200 mL/min — Example 4-3 - Flow rate of material solution: 250 mL/min — Example 4-4 - Flow rate of phosgene solution: 162 mL/min — Example 4-4 - Flow rate of material solution: 1.5 mL/min — Example 5-1 - Flow rate of phosgene solution: 1.08 mL/min — Example 5-1 - Microreactor temperature: 0° C. — Example 5-1 - Microreactor temperature: 20° C. — Example 5-2 - Microreactor temperature: 40° C. — Example 5-3 - Microreactor temperature: 60° C. — Example 5-4 - Microreactor temperature: 60° C. — Example 5-5 - GPC monomer area: 64.4% — Table 3, Example 4-1 - CF number: 0.88 N — Table 3, Example 4-1 - Concentration: 0.215 kg/L — Table 3, Example 4-1 - CF value: 4.1 N/kg — Table 3, Example 4-1 - Average number of repeating

units (n): 1.32 — Table 3, Example 4-1 - GPC monomer area: 86.1% — Table 3, Example 4-2 - CF number: 1.21 N — Table 3, Example 4-2 - Concentration: 0.251 kg/L — Table 3, Example 4-2 - CF value: 4.82 N/kg — Table 3, Example 4-2 - Average number of repeating units (n): 1.07 — Table 3, Example 4-2 - GPC monomer area: 91.4% — Table 3, Example 4-3 - CF number: 0.86 N — Table 3, Example 4-3 - Concentration: 0.14 kg/L — Table 3, Example 4-3 - CF value: 6.14 N/kg — Table 3, Example 4-3 - Average number of repeating units (n): 1.07 — Table 3, Example 4-3 - GPC monomer area: 96.5% — Table 3, Example 4-4 - CF number: 0.87 N — Table 3, Example 4-4 - Concentration: 0.23 kg/L — Table 3, Example 4-4 - CF value: 3.78 N/kg — Table 3, Example 4-4 - Average number of repeating units (n): 1.06 — Table 3, Example 4-4 - GPC monomer area: 96.2% — Table 3, Example 4-5 - CF number: 1.76 N — Table 3, Example 4-5 - Concentration: 0.32 kg/L — Table 3, Example 4-5 - CF value: 5.52 N/kg — Table 3, Example 4-5 - Average number of repeating units (n): 1.1 — Table 3, Example 4-5

**Relevant Experimental Evidence** - Example 1-1: Showed bischloroformate formation from biphenol/phosgene/TEA in dichloromethane at 13-16° C. - Example 1-3: Demonstrated triphosgene-based synthesis under low-temperature staged addition. - Experimental Example 3-1: Produced a polycarbonate oligomer with reduced viscosity 0.94. - Example 4-1: Showed microreactor flow synthesis with 50 mL/min material feed and n = 1.32. - Example 5-1: Demonstrated that 0° C. microreactor operation is feasible for bischloroformate formation.

**Technical Relevance** This patent is relevant as an oligomer-control analogue: it shows how stoichiometry, flow rate, and temperature can be used to produce low-repeating-unit products with controlled molecular size. The same process-control logic is applicable to low-MW NBR manufacturing, especially where oligomeric or liquid-rubber behavior is desired.

#### Patent 4

- Patent Number: US9434846B2
- Patent Title: Anti-settling and thickening compositions and methods for using same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: This patent discloses emulsion polymerization of HASE thickener polymers from carboxylic acid monomers, nonionic monomers, and hydrophobic associative monomers, followed by chaser treatment to consume residual monomer and reduce VOCs. The process uses nitrogen purging, elevated temperature, ammonium persulfate initiation, and surfactant-resistant formulation testing. - Nitrogen purge: maintained throughout the run — Example 1 - Reaction temperature: about 80° C. — Example 1 - Reaction hold time: over approximately 3 hours — Example 1 - Additional hold time: 30 minutes — Example 1 - Chaser addition time: 30 minutes — Example 1 - Surfactant: RHODAPEX AB20 —

Example 1 - Initiator: ammonium persulfate — Example 1 - Hydrophobic associative monomer structure: alkyl alkoxyate with N in the range 8 to 10 and M in the range 18 to 22 — Example 1 - Formulation mass: 108 g binder + 61 g water — Example 2 - Surfactant addition: 1 g IGEPAL CO887 — Example 2 - Surfactant addition: 2 g IGEPAL CO887 — Example 2 - Surfactant addition: 3 g IGEPAL CO887 — Example 2 - Viscosity measurement temperature: 25° C. — Example 2 - Stain mass: 150 g — Example 3 - Polymer addition: 2.0 g — Example 3 - Stain pH: 8.73 — Example 3 - Stability time: 168 hours — Example 3 - Stain separation time without polymer: within 5 hours — Example 3 - KU viscosity: 48.2 — Example 3, Sample 3 - KU viscosity: 51.5 — Example 3, Sample 4 - KU viscosity: 51.8 — Example 3, Sample 5 - ICI: 0.15 — Example 3, Samples 3-5 - Zahn cup time: 17.0 sec — Example 3, Sample 3 - Zahn cup time: 21.0 sec — Example 3, Sample 4 - Zahn cup time: 23.0 sec — Example 3, Sample 5 - Polymer weight average molecular weight: 1,000 to 1,200,000 g/mol — Claim 2 - Polymer molecular weight: 750,000 to 5,000,000 g/mol — Claim 3 - Coating viscosity: less than about 200 KU — Claim 4

**Relevant Experimental Evidence** - Example 1: Demonstrated HASE emulsion polymerization at 80° C. with ammonium persulfate and chaser chemistry to reduce residual monomer. - Example 2: Showed that thickener formulations retained KU viscosity after surfactant addition. - Example 3: Demonstrated improved stain stability for 168 hours with only modest viscosity increase. - Example 3: Showed untreated stain separated within 5 hours, while polymer-treated stains remained homogeneous. - Example 1: Illustrated use of hydrophobic associative monomer with N = 8-10 and M = 18-22.

**Technical Relevance** This patent is indirectly relevant because it demonstrates emulsion polymerization control, residual monomer consumption, and viscosity management in a rubber-like polymer system. While not NBR, it provides useful precedent for controlling polymer architecture and low-viscosity performance in aqueous polymerization processes.

## 3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10167348B2 and US9434846B2 both rely on solution/emulsion polymerization with explicit control of initiator chemistry and temperature, but US10167348B2 emphasizes anionic living control and molecular-weight tuning, whereas US9434846B2 emphasizes emulsion thickener performance and residual monomer conversion.
- US10167348B2 provides the clearest molecular-weight control levers for low-MW targets: monomer-to-activator ratios of about 1000:1 or 100:1, temperatures from about 23° C. to -78° C., and solvent-dependent Mn values from about 2,000 to 2,000,000 daltons.
- US7683134B2 uses a continuous melt extrusion route rather than solution polymerization; molecular weight is influenced by screw speed, peroxide feed rate, and maleic anhydride feed, with Mn values spanning about 30,000 to 71,000 in the examples.
- US8470957B2 is an oligomer-control patent rather than a rubber patent, but it shows that low repeating-unit products can be obtained by tight stoichiometric and flow control, with average



repeating units  $n$  around 1.06 to 1.32 in the examples.

- Across the set, the most transferable low-MW design principles are: controlled initiator/activator ratio, low-temperature reaction control, continuous or staged monomer addition, and post-reaction quenching or termination to freeze molecular weight.
- None of the patents directly disclose NBR monomer ratios of butadiene to acrylonitrile or NBR-specific chain-transfer agents; therefore, the evidence is best interpreted as process analogues for molecular-weight control rather than direct NBR recipes.

## 4. CONCLUSION

The patent set does not include a direct low-MW NBR synthesis disclosure, but it does provide strong process analogues for controlling molecular weight and viscosity in polymer manufacture. The most relevant evidence supports solution polymerization with tight monomer-to-initiator control, low or subambient temperatures when needed, staged monomer addition, and rapid quenching to arrest growth. For a low-MW NBR program, the best-supported synthesis strategy from this evidence would be controlled solution polymerization with carefully limited initiator or chain-transfer activity, solvent selection that moderates heat and viscosity, and process conditions that avoid runaway conversion and broad molecular-weight distributions. Continuous or semi-continuous operation with in-line monitoring would be preferable if scale-up and reproducibility are priorities. Based on the disclosed examples, the most transferable parameter set is a low-temperature, solvent-based controlled radical or anionic process with explicit feed-ratio management and immediate termination once the desired  $M_n$  window is reached.

## 5. REFERENCES

- US10167348B2 | Solution polymers formed from methylene malonate monomers, polymerization, and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10167348B2/en>
- US7683134B2 | Maleated high acid number high molecular weight polypropylene of low color | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7683134B2/en>
- US8470957B2 | Method for producing bischloroformate compound, polycarbonate oligomer having ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8470957B2/en>
- US9434846B2 | Anti-settling and thickening compositions and methods for using same | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9434846B2/en>